PHOENICIAN CAPITAL · LOCAL-ENGINE-INTERNET

Live demo
Search + DeepSeek Flash

A short paper you can send. How the service works, what we tested on 9 September 2026, and the live numbers DeepSeek Flash produced after searching and reading Apple's own filings. No cloud writing models. Same Flash you will run in the datacenter.

Engine: http://127.0.0.1:8090

Model: deepseek-v4-flash (DeepSeek API today · local Brain tomorrow)

Search: SerpAPI + Tavily

Date of run: 9 September 2026, 10:50–10:55 UTC

Internals (full code map): Local-LLM-Internet-Engine-Method.pdf

1. How it works

Company apps already talk to DeepSeek with `POST /v1/chat/completions` (messages, tools, tool_calls). This service sits on port `8090` and speaks that same DeepSeek / vLLM wire format. It does not load model weights. It searches the web, downloads pages, and — in chat — sits in front of DeepSeek Flash.

```
Analyst / PI / portfolio app
        │
        ▼
local-engine-internet  :8090
   ├── search + fetch     SerpAPI, Tavily  (Brave later)
   └── chat + tools       DeepSeek Flash
                                │
                    today:  api.deepseek.com
                    later:  ai-router :8080  (same model name)
```

Mode 1 — the pipeline already decided to search. `POST /v1/search`, `/v1/fetch`, `/v1/research`. No model in the hop.

Mode 2 — the user sends a question. Flash may call `web_search` and `fetch_url`. The engine runs those tools and asks Flash again until it answers in prose. Sources stay labeled. If search is required and fails, the HTTP status is 424 — it does not invent numbers from training data.

When the datacenter Brain is up, change one setting: `UPSTREAM_LLM_BASE_URL=http://127.0.0.1:8080`. Apps keep pointing at `http://127.0.0.1:8090/v1`. You stop using Claude / GPT / Gemini / Perplexity as the writer. You still use search APIs (Google via SerpAPI, Tavily). Those are indexes, not models.

2. What we tested

Two batteries on the same machine, same engine process (restarted only to point chat at DeepSeek's API).

Check	What it proves	Result
Health / plugin	Service up, internet plugin on, model may choose when to search	**Pass**
SerpAPI + Tavily mix	Both indexes run in parallel; one failure does not kill the request	**Pass – 0 errors**
GET /search	SerpAPI-shaped JSON for existing scrapers	**Pass**
Fetch HTML	example.com and investor.apple.com	**Pass**
Fetch official 10-Q PDF	Parser reads Apple's IR PDF (income statement)	**Pass**
Fail-closed (no Brain)	Chat while :8080 is down	**502 – search still works; no invented answer**
Tool-call canary (DeepSeek API)	Flash emits structured tool_calls	**Pass**
Mode 2, policy required	Flash must search, then answer with URLs	**200 in 77.6s**
Mode 2, policy off	"What is 2+2?" stays offline	**"4" – no tools, 1.1s**
pytest (71 tests)	No live keys; locks mix, plugin, fail-closed, fetch ladder	**Pass (run separately)**

Brave was not configured. It is optional — the mix runs without it. Tavily returned 403 earlier in the day and 200 on retry; the mix skipped the failure and kept SerpAPI.

3. Live Mode 1 – search and fetch

Ran 10:50–10:51 UTC. No model. Query used for the clean hit list:

```
AAPL 10-Q "June 27, 2026" OR aapl-20260627 site:sec.gov OR site:q4cdn.com
```

	Value
HTTP	200
Time	3.7 seconds
Providers used	serpapi, tavily
Errors	none
Hits	8
Cross-confirmed URL (both indexes)	https://www.sec.gov/Archives/edgar/data/320193/000032019326000020/aapl-20260627.htm

Tavily also returned Apple's 30 July 2026 newsroom / 8-K and the FY26 Q3 statements PDF. A looser query ("Apple Inc latest 10-Q…") still returned both providers but mixed in junk (wrong CIK, unrelated video). The engine does not hide bad hits — Flash (or an analyst) must weigh them. That is intentional.

Official PDF we fetched ourselves

`POST /v1/fetch` of Apple's IR copy of the Q2 FY2026 10-Q (quarter ended 28 March 2026), 3.8s, httpx, no browser:

https://s2.q4cdn.com/470004039/files/doc_earnings/2026/q2/filing/10Q-Q2-2026-as-filed.pdf

Line extracted from the PDF	Q2 FY26	Prior-year quarter
Total net sales	111,184	95,359
Operating income	35,885	29,589
Net income	29,578	24,780
Diluted EPS	2.01	1.65

These figures came out of the PDF parser, not a model. Q2 ≠ Q3. The later chat answer is Q3 (June quarter). Keeping them separate is the point of labeled sources.

4. Live Mode 2 – DeepSeek Flash with tools

Ran 10:54 UTC. Engine on :8090. Flash at `https://api.deepseek.com`, model `deepseek-v4-flash`. Same name you will load on vLLM.

Prompt we sent

```
Apple Inc (AAPL) most recently reported quarterly revenue, operating
income, and diluted EPS. Prefer the official 10-Q or earnings release
on SEC EDGAR or investor.apple.com. Return the filing URL.
```

Body extra: `phoenician_web.policy: "required"` — Flash must complete a successful search or the engine returns 424.

	Value
HTTP	**200**
Wall time	77.6 seconds
Tool rounds	4 × web_search, 4 × fetch_url
Labeled sources	26
Tokens (all rounds)	70,782
Canary	brain_tool_calls_supported = true

What Flash called

1. `web_search` — Apple latest quarterly results / 10-Q
2. `web_search` — Q3 comparables / press release
3. `web_search` — apple.com newsroom, 27 June 2026
4. `fetch_url` — SEC EDGAR browse for CIK 0000320193, type 10-Q
5. `fetch_url` — filing folder 0000320193-26-000020
6. `fetch_url` — FY26_Q3_Consolidated_Financial_Statements.pdf (Apple Newsroom)
7. `fetch_url` — EDGAR index for that accession
8. `web_search` — "Apple Reports Third Quarter Results" 2026

What Flash answered

Q3 FY2026, quarter ended 27 June 2026, reported 31 July 2026. From Apple's condensed consolidated statements (unaudited), millions except per share:

Metric	Q3 FY2026	Q3 FY2025
Total net sales (revenue)	**109,417**	94,036
Operating income	**35,695**	28,202
Net income	**29,789**	23,434
Diluted EPS	**2.02**	1.57

URLs it cited:

- https://www.apple.com/newsroom/pdfs/fy2026q3/FY26_Q3_Consolidated_Financial_Statements.pdf
- https://www.sec.gov/Archives/edgar/data/320193/000032019326000020/ (accession 0000320193-26-000020)

Control — no web

Same engine, `policy: off`, question “What is 2+2? Reply with the number only.” Answer: **4**. No tool trace. 1.1 seconds. The plugin can stay on while a request stays offline.

5. What this means for the datacenter

```
today     engine :8090  →  api.deepseek.com     →  deepseek-v4-flash
later     engine :8090  →  ai-router :8080      →  same Flash, local weights
```

vLLM flags the Brain needs so Flash emits structured `tool_calls` (same JSON DeepSeek's own API uses):

```
--tokenizer-mode deepseek_v4
--tool-call-parser deepseek_v4
--enable-auto-tool-choice
--reasoning-parser deepseek_v4
```

One cloud quirk we already handled: DeepSeek V4 thinking mode rejects a forced `tool_choice` (the canary and `policy: required`). The engine sends `thinking: {type: disabled}` only when the upstream host is `api.deepseek.com`. Local vLLM is unchanged.

Not in this run: Brave key, self-hosted SearXNG, the metal Brain on :8080. None of those block the path above.

6. How to re-run

```
python -m engine                          # :8090
python scripts/check_search.py --engine   # Mode 1, no model
python scripts/smoke.py                   # Mode 2, needs Flash up

# Same prompt as §4
curl -s http://127.0.0.1:8090/v1/chat/completions \
  -H 'Content-Type: application/json' \
  -d '{
    "model": "deepseek-v4-flash",
    "messages": [{"role":"user","content":"Apple Inc (AAPL) most recently reported quarterly revenue,
operating income, and diluted EPS. Prefer the official 10-Q or earnings release on SEC EDGAR or
investor.apple.com. Return the filing URL."}],
    "phoenician_web": {"policy": "required"}
  }'
```

Look for `phoenician_tool_trace` (web_search / fetch_url) and `phoenician_sources` with real http URLs. Toggle: http://127.0.0.1:8090/ui · OpenAPI: http://127.0.0.1:8090/docs.

This paper is the live run. The long internals manual (every file under engine/) is Local-LLM-Internet-Engine-Method.pdf. Neither document contains API keys.